GEORGE BOUFIS

ASSIGNMENT 2 (Probability and Statistics for data analysis)

Exercise 1)

Data Inspection and Descriptive Statistics

Initially, the dataset was imported into R, consisting of 30 observations divided equally into three experience levels : Novice, Advanced and Proficient (n = 10 per group). The following table summarizes the central tendency and dispersion of the recall scores across the entire sample.

Min.Score = 22.10

1st Quantile = 37.55

Median = 44.35

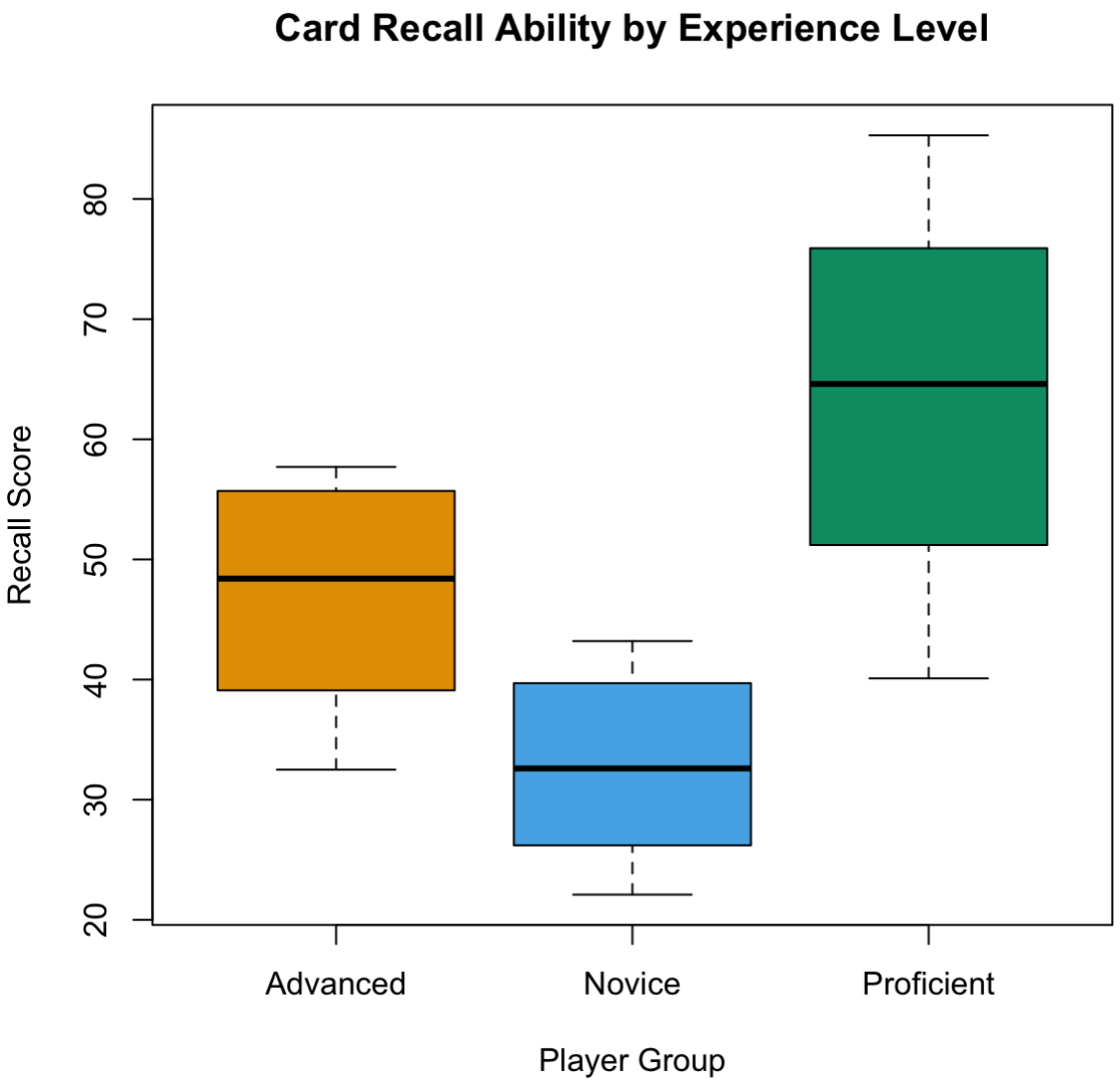
Mean = 47.91

3rd Quantile = 56.27

Max.Score = 85.30

The summary statistics reveal a wide range of scores, from minimum of 22.10 to maximum of 85.30, this significant gap between min.score and max.score suggest high variability.While the max.score might appear as an outlier, it likely reflects the superior performance expected in Proficient group. The mean (47.91) is slightly higher than the median (44.35), indicating a positive skewness in the data distribution. This is likely driven by the higher performance observed in the Proficient group. The balance designed ensures that our subsequent ANOVA analysis will be robust.

To visualize the distribution scores across the three categories a boxplot was generated.



Observations from the boxplot :

There is a clear, visible progression in recall scores across the three levels. The Novice group has the lowest median score, followed by the Advanced group, while the Proficient group demonstrates the highest performance.
Also, the median lines (black lines inside the boxes) for each group are located at different heights and do not overlap. Specifically, the median for the proficient group is higher than the entire Interquantile Range (IQR) of both novice and advanced groups.
According to the variability (spread), the vertical size of the boxes indicates that the proficient group has much higher variability in scores compared to other groups.This means that while experts generally perform better, their scores are more spread out.
The dashed lines show the full range of the data. The proficient group reaches the highest score (85.30), while the novice group reaches the lowest score (22.10).

These visual findings suggest that experience level is a significant factor in recall ability, providing justification for the subsequent One-Way ANOVA test.

b) **One-way ANOVA results and hypothesis testing**

To formally assess whether experience levels impact card recall ability, a One-Way ANOVA was conducted.The null hypothesis (Ho) states that all the groups have the same mean ($\mu 1 = \mu 2 = \mu 3$) while the alternative hypothesis suggests that at least one group mean is different.

Table of ANOVA :

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Group	2	4777	2389	18.37	9.21e-06
Residuals	27	3511	130		

The ANOVA results reveal a highly significant effect of experience level on recall scores ($F(2, 27) = 18.37$, $p<0.001$).Since the p-value(9.21e-06) is lower than the predetermined significance level a = 0.05, then we reject the null hypothesis.
The F-statistic indicated that the variance between the groups is significantly larger than the variance within the groups confirming that the observed differences in the boxplot are not due to random chance.However, while ANOVA tells us at least one difference exists, it does not specify which groups differ.Therefore a post-hoc analysis is required.

c) **Pairwise comparisons and Methodological Validity**

Following ANOVA, pairwise t-tests were performed to identify specific differences between the groups. The results, without p-value adjustment are shown below :

Comparison	p-value (unadjusted)
Advanced vs. Novice	0.0119
Proficient vs. Advanced	0.0024
Proficient vs. Novice	1.9e-06

Although all three p-values are below the 0.05 threshold, it is not statistically valid to use these specific results to draw final conclusions. When testing the three pairs simultaneously (Novice-Advanced, Novice-Proficient, and Advanced-Proficient) at a 5% significance level, the probability of committing at least one **Type I Error** increases. Without an adjustment method, the actual risk of a false positive result across the entire experiment is significantly higher than the intended 5%.

d) **Post-hoc Analysis: Tukey Honestly Significant Difference (HSD)**

To properly identify which groups differ maintaining a 95% confidence level,a Tukey HSD test was performed. This method adjusts the p-values to account for multiple comparisons, providing a more reliable basis for decision making than unadjusted t-tests.

Comparison	Difference (diff)	Lower Bound (lwr)	Upper Bound (upr)	Adjusted p-value (p adj)
Novice - Advanced	-13.75	-26.394476	-1.105524	**0.0310169**
Proficient - Advanced	17.10	4.455524	29.744476	**0.0065079**

Proficient - Novice	30.85	18.205524	43.494476	**0.0000054**

Based on the adjusted p-values ($p<0.05$ for all pairs) we conclude that every group is significantly different from the others :

Novice vs Advanced : Advanced players scored significantly higher than novices (diff = 13.75, p = 0.0310169). The confidence interval [-26.394476, -1.105524] does not include zero (reject Ho), confirming the significance.

Proficient vs. Advanced : Proficient players outperformed advanced ones (diff = 17.10, p = 0.0065079). Also here we can see that the confidence interval does not include zero, so we tend to reject Ho.

Proficient vs. Novice: The largest difference was observed here, since diff = 30.85 and $p < 0.001$ showing that proficient players have a vastly superior recall ability compared to novices.

Conclusion :
The statistical evidence suggests that the ability to recall cards is directly and significantly related to a player's experience level. Each progression in experience (from Novice to Advanced and from Advanced to Proficient) results in a statistically significant improvement in recall scores

Exercise 2)

1. Data Import and Descriptive Statistics

The objective of this study is to examine if alcohol consumption relates to partner selection and whether this relationship is moderated by gender. The dataset consists of 48 students(n = 48), balanced with 24 males and 24 females. Participants were divided into three equal groups based on alcohol intakes : None, 2 pints, 4 pints.

Statistic	Value
Minimum Score	20.00
1st Quartile	53.75
Median	60.00
Mean	58.33
3rd Quartile	66.25
Maximum Score	85.00

The initial inspection of the data shows a total sample size of 48 participants, with 16 individuals assigned to each alcohol treatment group. The attractiveness scores range from a minimum of 20 to a maximum of 85, with a mean score of 58.33 and a median of 60.00. The proximity of the mean to the median suggests that the overall distribution of scores is relatively symmetric. The balanced experimental design, featuring equal numbers of males and females across the alcohol levels, provides a strong foundation for conducting a **Two-Way ANOVA** to test for interaction effects.

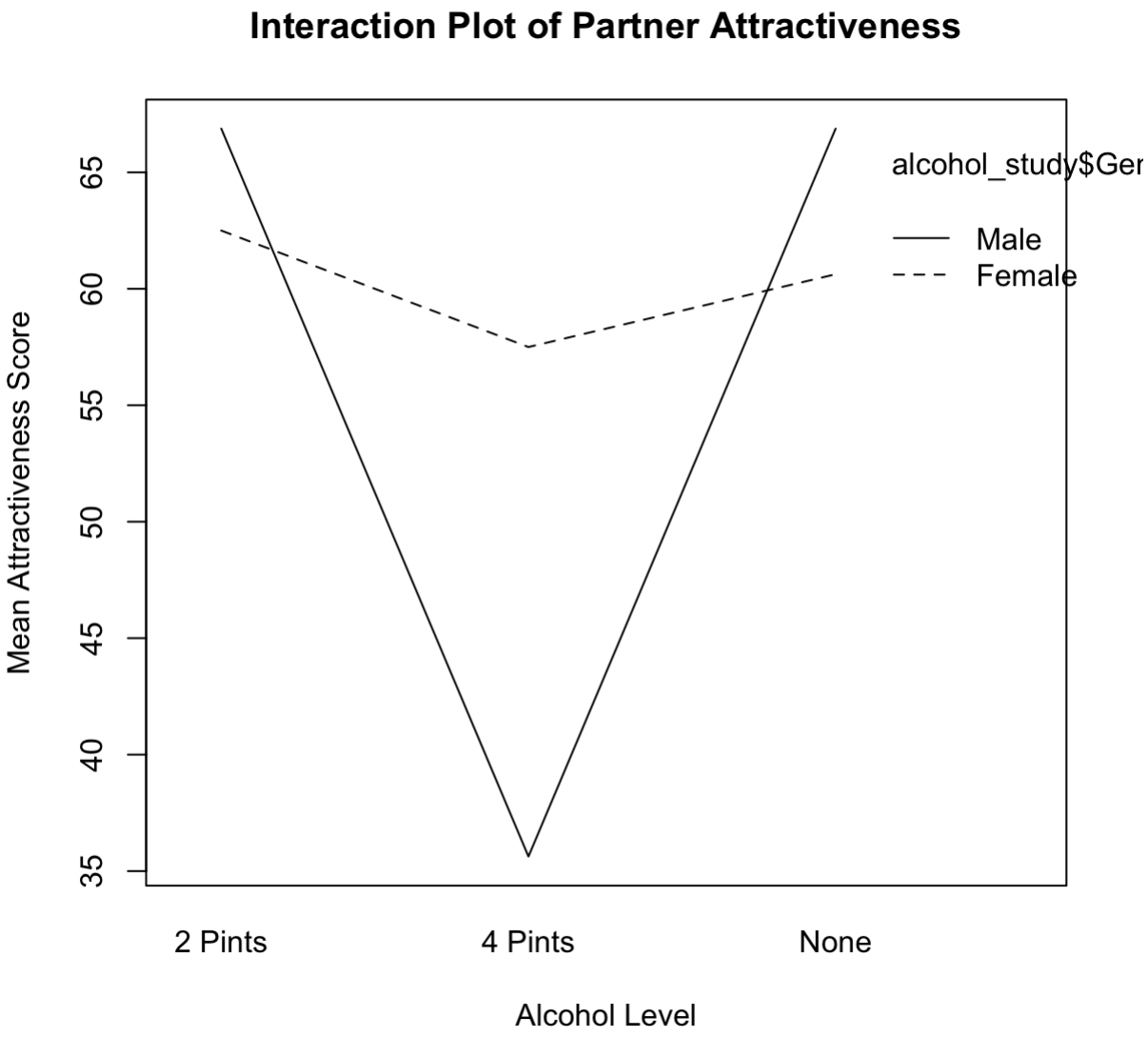
To further inspect the data, we calculated the mean and standard deviation for each combination of Gender and Alcohol level.

Alcohol Level	Gender	Mean Score	Std. Deviation
None	Female	60.62	4.95

None	Male	66.87	10.33
2 Pints	Female	62.50	6.55
2 Pints	Male	66.87	12.52
4 Pints	Female	57.50	7.07
4 Pints	Male	**35.62**	10.84

The group-level statistics suggest a potential interaction effect between Gender and Alcohol consumption. While female participants maintained relatively stable attractiveness ratings across all alcohol levels, male participants showed a dramatic decrease in their ratings after consuming 4 pints of beer (dropping to a mean of 35.62). This sharp decline in the male group, compared to the relative stability in the female group, indicates that alcohol may affect partner selection differently depending on gender.

The **Interaction Plot** reveals non-parallel lines, which is a classic indicator of a significant interaction effect. The dashed line (Females) remains relatively horizontal across all alcohol levels, indicating stable judgment. In contrast, the solid line (Males) shows a sharp decline at the '4 Pints' level. This visual evidence explains why the interaction term in our ANOVA was highly significant ($p < 0.001$) alcohol consumption drastically reduces attractiveness ratings, but only for male participants at high consumption levels.



2) To formally analyze these data, we propose a **Two-Way ANOVA model with interaction**. This model is appropriate because we have one continuous response variable (Attractiveness Score) and two categorical independent variables (Alcohol and Gender).
The statistical model is defined as:
$Y_{ijk} = \mu + a_i + \beta_j + (a\beta)_{ij} + \varepsilon_{ijk}$
Where :
Y_{ijk} : is the attractiveness rating for an individual.
μ : is the overall mean attractiveness score.
a_i : is the main effect of the **Alcohol** level (None, 2 pints, 4 pints).
β_j : is the main effect of **Gender** (Male, Female).
$(a\beta)_{ij}$: represents the **Interaction** effect between Alcohol and Gender.
ε_{ijk} : is the random error term.

3) To determine if partner selection depends on alcohol consumption, gender, or their interaction, a Two-Way ANOVA was performed at a 5% significance level.

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Alcohol	2	3332	1666.1	20.065	7.65e-07 ***
Gender	1	169	168.7	2.032	0.161
Alcohol:Gender	2	1978	989.1	11.911	7.99e-05 ***
Residuals	42	3487	83.0		

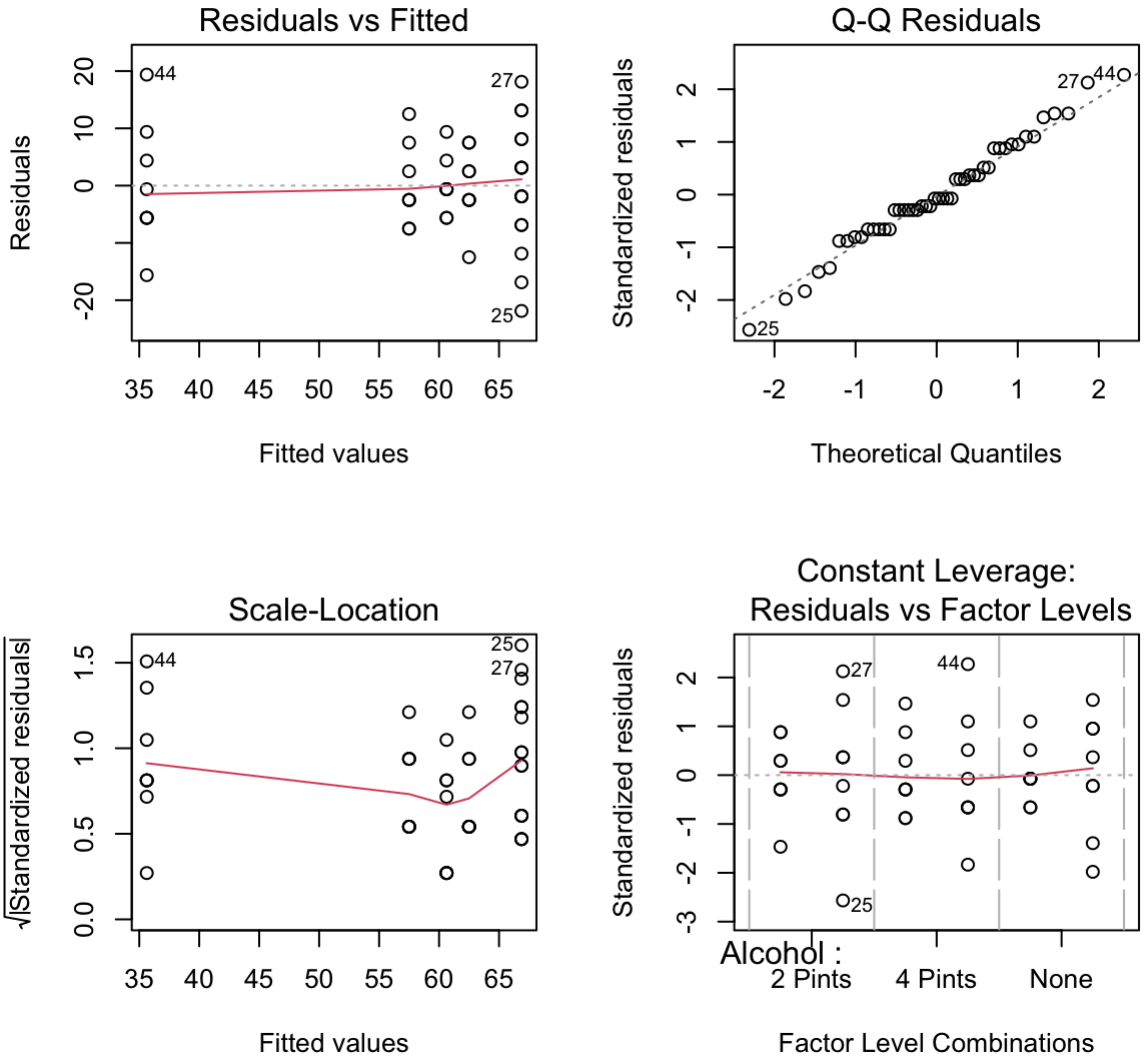
The interaction between alcohol and gender is highly significant ($F(2,42) = 11.911$, $p<0.001$).This is the most critical finding as it indicates that the effect of alcohol on attractiveness ratings is not the same for both genders. Because this interaction is significant, we must interpret the main effects of Alcohol and Gender.

Main effect of Alcohol : There is a significant main effect of alcohol consumption ($F(2,42) = 20.065$, $p<0.001$). This confirms that the amount of alcohol consumed generally affects how participants rate partner attractiveness.

Main effect of Gender : The main effect of gender is not statistically important ($F(1,42) = 2.032$, $p = 0.161>0.05$). This means that, on average (across all alcohol levels), there is no evidence that one gender provides consistently higher or lower ratings than the other.

4)
The **Shapiro-Wilk normality test** was performed on the model's residuals. The resulting p-value was **0.6643**. Since this is significantly higher than the a = 0.05 threshold, we fail to reject the null hypothesis, confirming that the residuals follow a **normal distribution**.

Levene's test was used to assess whether the variances across the different groups (AlcoholxGender) are equal. The test yielded a p-value of **0.2351**. Because $p > 0.05$, the assumption of **homoscedasticity** (equal variances) is satisfied.



Residuals vs Fitted :

This plot checks the assumption of **linearity** and **constant variance** (homoscedasticity).

a) **Observation:** The points are randomly scattered around the horizontal dashed line at zero without forming any distinct shape (like a funnel or a curve).

b) **Conclusion:** This supports the assumption of constant variance across all groups, consistent with the non-significant result of the Levene's test.

Normal Q-Q :

This plot evaluates the **normality** of the residuals.

a) **Observation:** Most of the data points fall almost perfectly along the straight diagonal line. While there are slight deviations at the extreme ends (labeled points 25, 27, 44), they are not severe enough to suggest a violation.

b) **Conclusion:** The residuals are approximately normally distributed, as previously confirmed by the Shapiro-Wilk test (p=0.6643).

Scale-Location :

This is another way to check for **homoscedasticity**.

a) **Observation:** The red trend line is relatively flat, and the spread of standardized residuals does not change systematically with the fitted values.

b) **Conclusion:** This reinforces that the variance of the errors is stable.

Constant Leverage: Residuals vs Factor Levels :

This plot helps identify **influential observations** or outliers that could disproportionately affect the model results.

Observation: The points are grouped by the interaction levels (Alcohol:Gender). All points fall well within the expected limits, and no specific observation appears to have high leverage that would bias the ANOVA coefficients.

The **Residuals vs Fitted** and **Scale-Location** plots indicate that the assumption of homoscedasticity is met, as the residuals exhibit constant spread across the range of fitted values. Furthermore, the **Normal Q-Q** plot demonstrates that the residuals follow a normal distribution, with only minor deviations at the tails. Finally, the **Residuals vs Factor Levels** plot shows no evidence of highly influential outliers. Together with the Shapiro-Wilk and Levene's tests, these plots provide strong evidence that all modelling assumptions are satisfied, and the findings regarding alcohol and gender are statistically valid.

5)
In this experiment, we looked at how drinking beer affects the way people judge the attractiveness of others and whether this effect is different for men and women.

The Beer Effect is Real, but Specific**:** Our study confirms that drinking alcohol changes how people perceive physical beauty, but it doesn't happen the same way for everyone.

Moderate Drinking (2 pints): When participants drank a small amount of beer (2 pints), their judgment remained almost identical to when they were sober. Both men and women were just as picky about their partner's looks as they were with non-alcoholic drinks.

High Consumption (4 pints): A major shift occurred after 4 pints of beer. We observed that while women's standards stayed relatively stable, **men's standards dropped significantly**. On average, men who had 4 pints gave much higher attractiveness ratings to partners that they would normally find less attractive.

Exercise 3)

1) To evaluate management's belief that technical documents require more reading time than general ones, a Welch Two-Sample t-test was conducted.

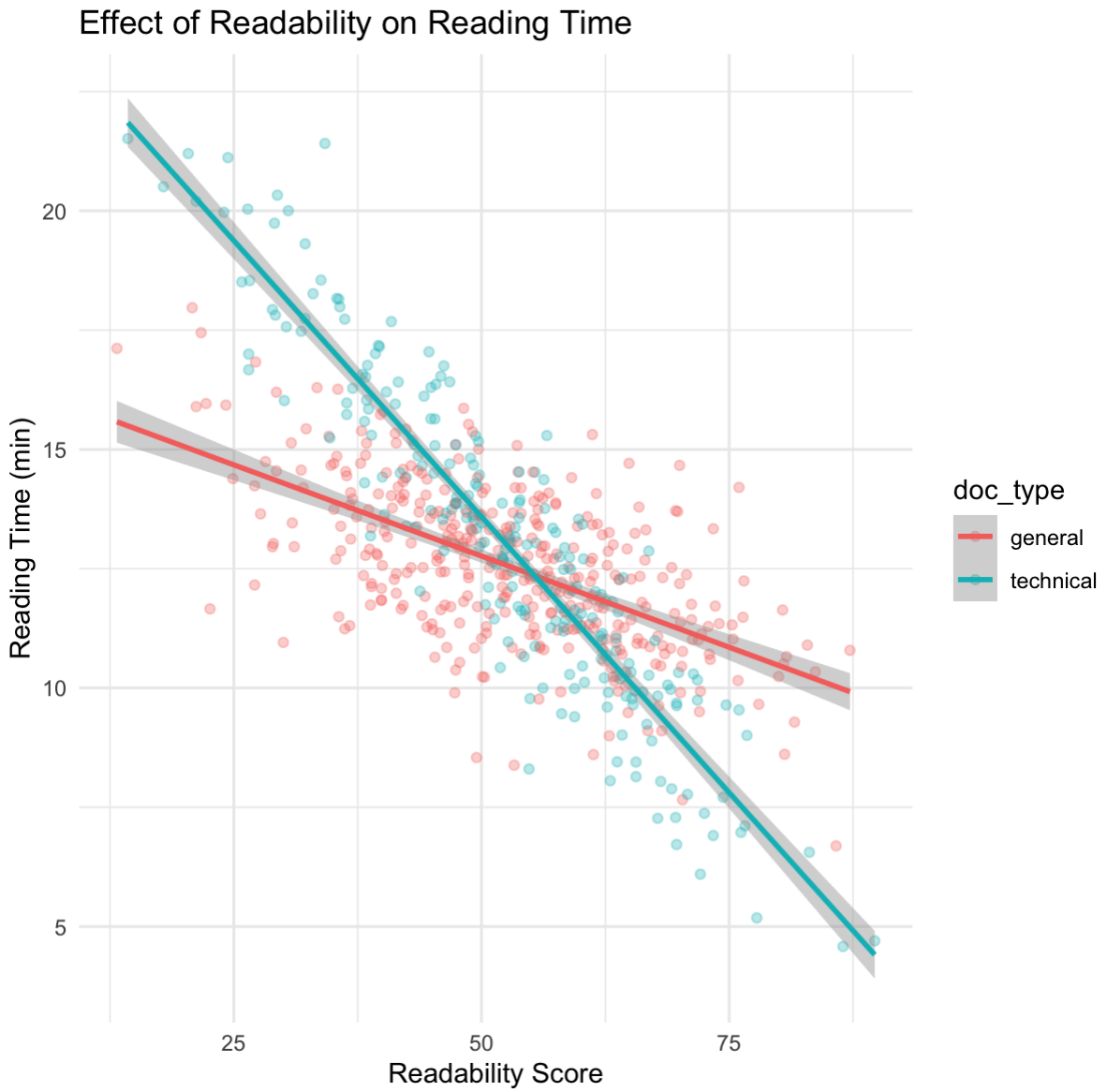
Results:

a) **Mean Reading Time (General):** 12.56 minutes
b) **Mean Reading Time (Technical):** 13.02 minutes
c) **p-value:** 0.0572

At a 5% significance level (a = 0.05), we **fail to reject the null hypothesis** (p = 0.0572 > 0.05). Although the sample mean for technical documents is slightly higher, there is not enough statistical evidence to conclude that a significant difference exists between the two types when considering reading time in isolation.

The marginal non-significance of this result suggests that the relationship is more complex. The simple comparison is likely confounded by other variables. For instance, if the word counts or complexity levels are similar across the two groups in this specific sample, the document type alone may not appear significant. This necessitates a multiple regression analysis to control for these additional features and uncover the true drivers of reading time.

2)



To investigate if the relationship between reading time and document features differs by category, we visualized the impact of the **Readability Score** across both document types.

The scatter plot reveals a strong negative correlation: as the readability score increases (meaning the text becomes easier), the reading time decreases for both groups. However, we observe a significant **interaction effect** evidenced by the non-parallel regression lines:

a) **General Documents (Red Line):** The slope is relatively shallow. This suggests that while easier text reduces reading time, the gain is moderate.
b) **Technical Documents (Blue Line):** The slope is much steeper. This indicates that technical documents are far more sensitive to readability. When a technical document is complex (low score), the reading time is significantly higher than a general document of similar complexity. Conversely, if a technical document is made very

clear (high score), the reading time drops more drastically than it does for general texts.

3)

Following the exploratory analysis, a multiple linear regression model was refined using a stepwise selection procedure. This approach identified the **minimal set of predictors** necessary to provide an adequate and parsimonious description of the reading time.

Model Specification and Justification: The final model includes main effects for nearly all variables and, crucially, several interaction terms with doc_type.

a) **Minimal Set of Predictors:** The model retains word_count, avg_sentence_length, avg_syllables_per_word, readability, image_count, doc_type, avg_paragraph_length, and metadata_score.

b) **Interaction Terms:** Several interactions (e.g., with paragraph length and metadata score) were retained, confirming that the impact of these structural features depends on whether the document is general or technical.

Word Count (β~0.0098) : Holding other factors constant, each additional word increases the predicted reading time by approximately **0.01 minutes**. This effect is highly significant ($p < 0.001$).

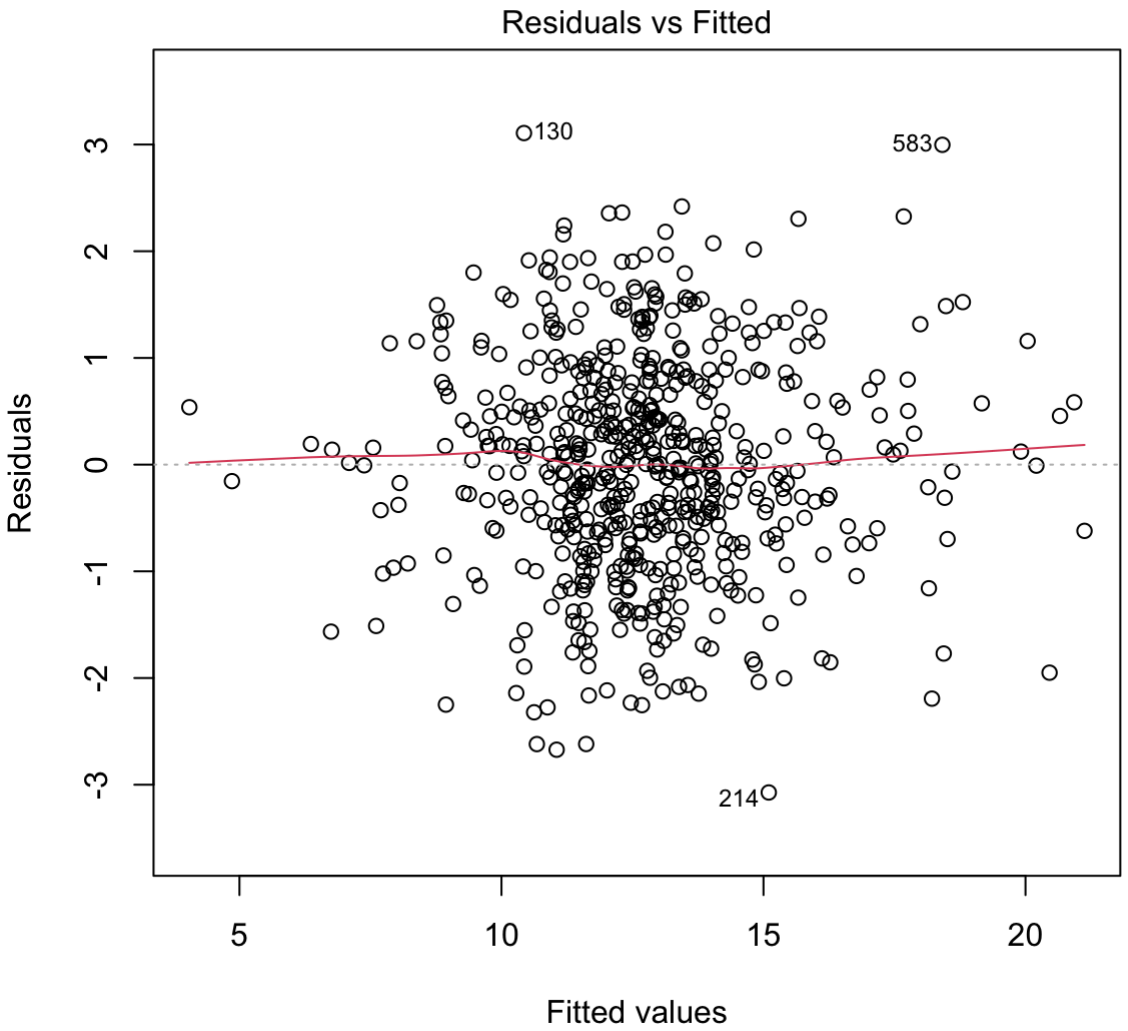
Image Count (β~ - 0.5598): Interestingly, for each additional image, the reading time **decreases by about 0.56 minutes**. This suggests that images may help readers process information more efficiently or act as substitutes for complex text.

Average Paragraph Length: This variable is significant as a main effect ($p = 0.027$) and also through its interaction with technical documents ($p = 0.028$). This indicates that the way paragraph structure affects reading speed differs between the two document types.

Metadata Score: While not significant for general documents, its interaction with technical documents ($p = 0.046$) suggests that high-quality metadata specifically aids the reading process in technical contexts.

Adjusted R-squared: 0.8284 This value indicates that **82.84%** of the total variability in document reading times is explained by our selected predictors and their interactions.

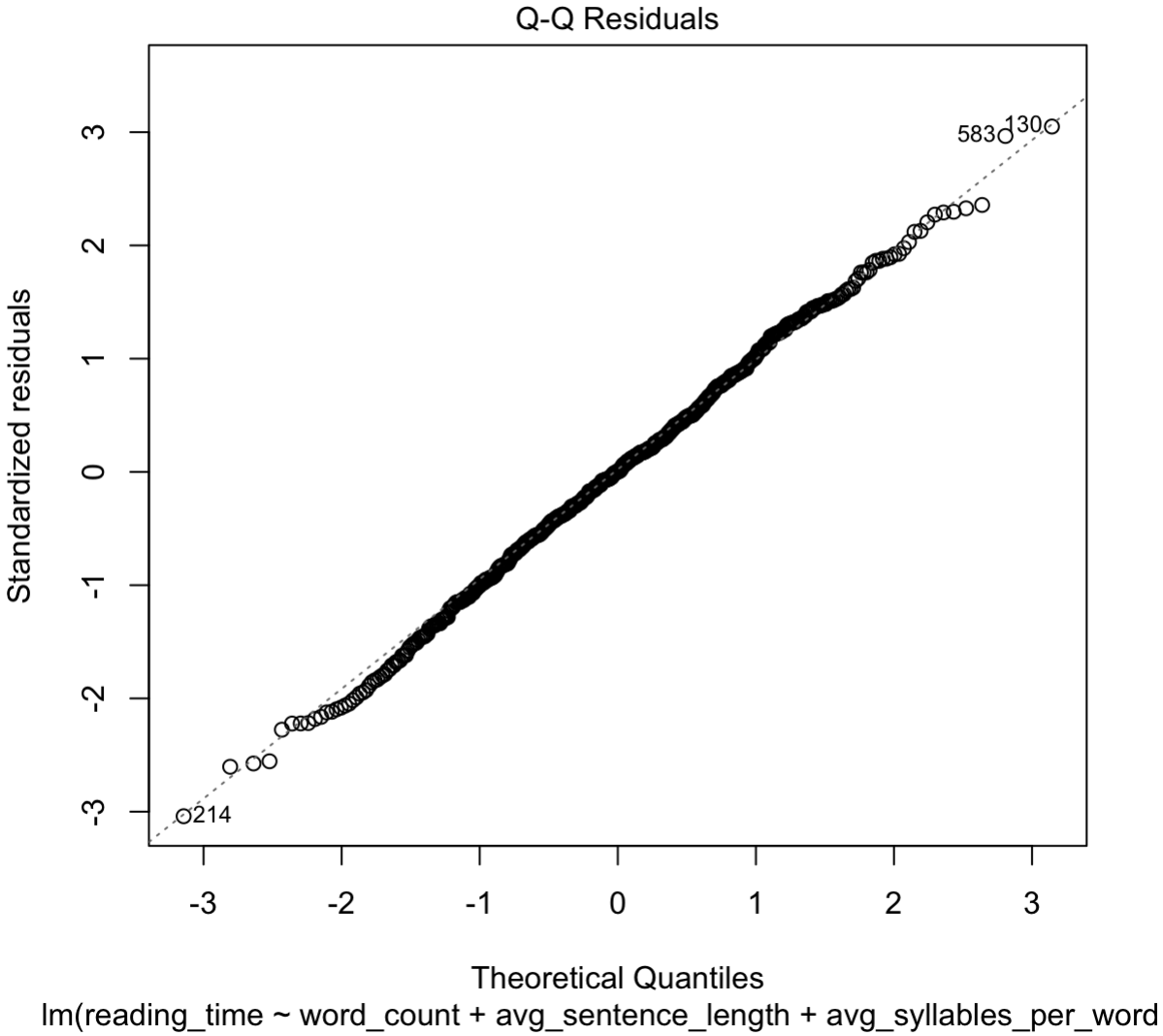
4)



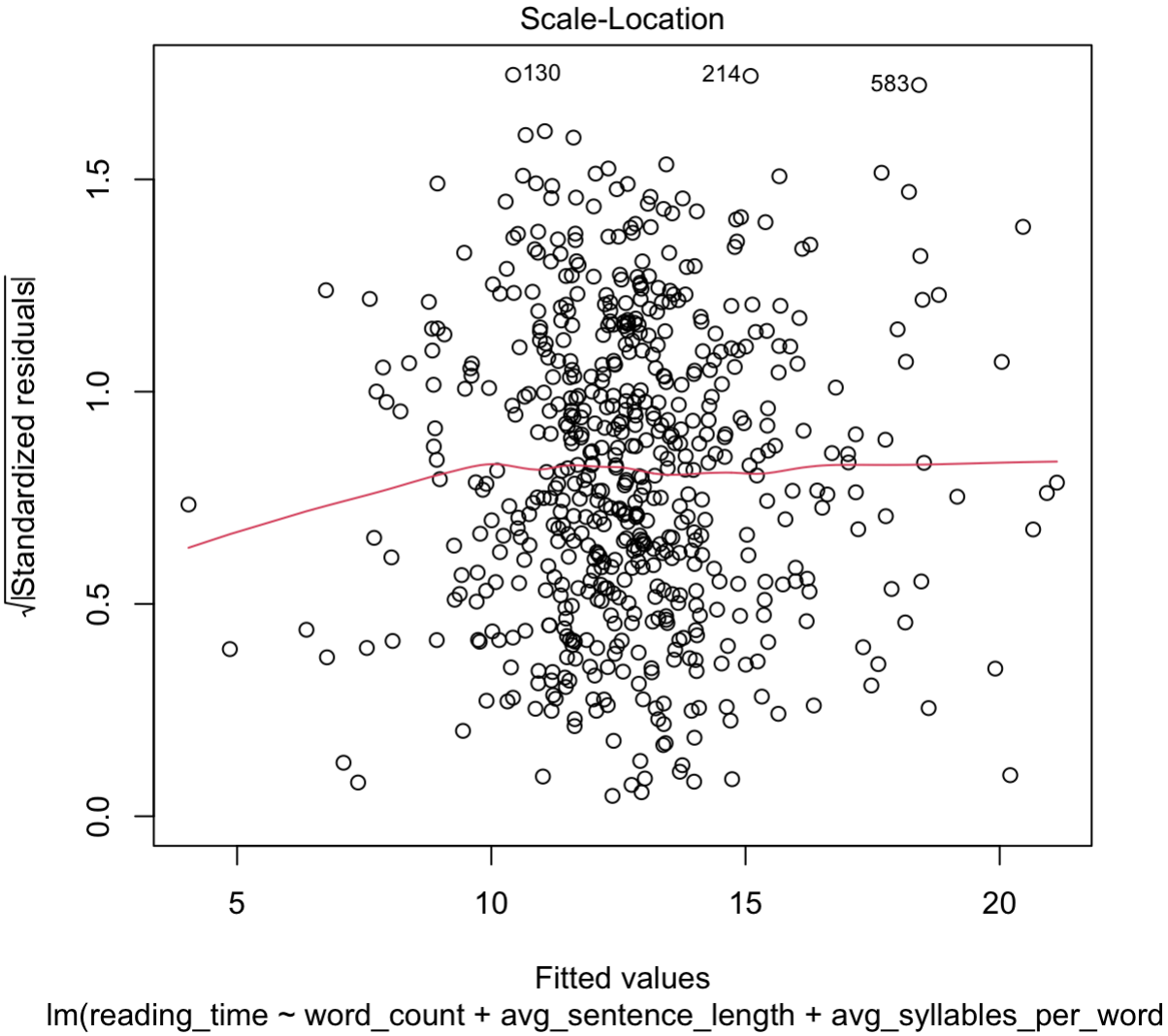
lm(reading_time ~ word_count + avg_sentence_length + avg_syllables_per_word

Linearity and Homoscedasticity (Residuals vs Fitted Plot): The **Residuals vs Fitted** plot (Figure X) is used to verify that the relationship between predictors and reading time is linear and that the error variance is constant.

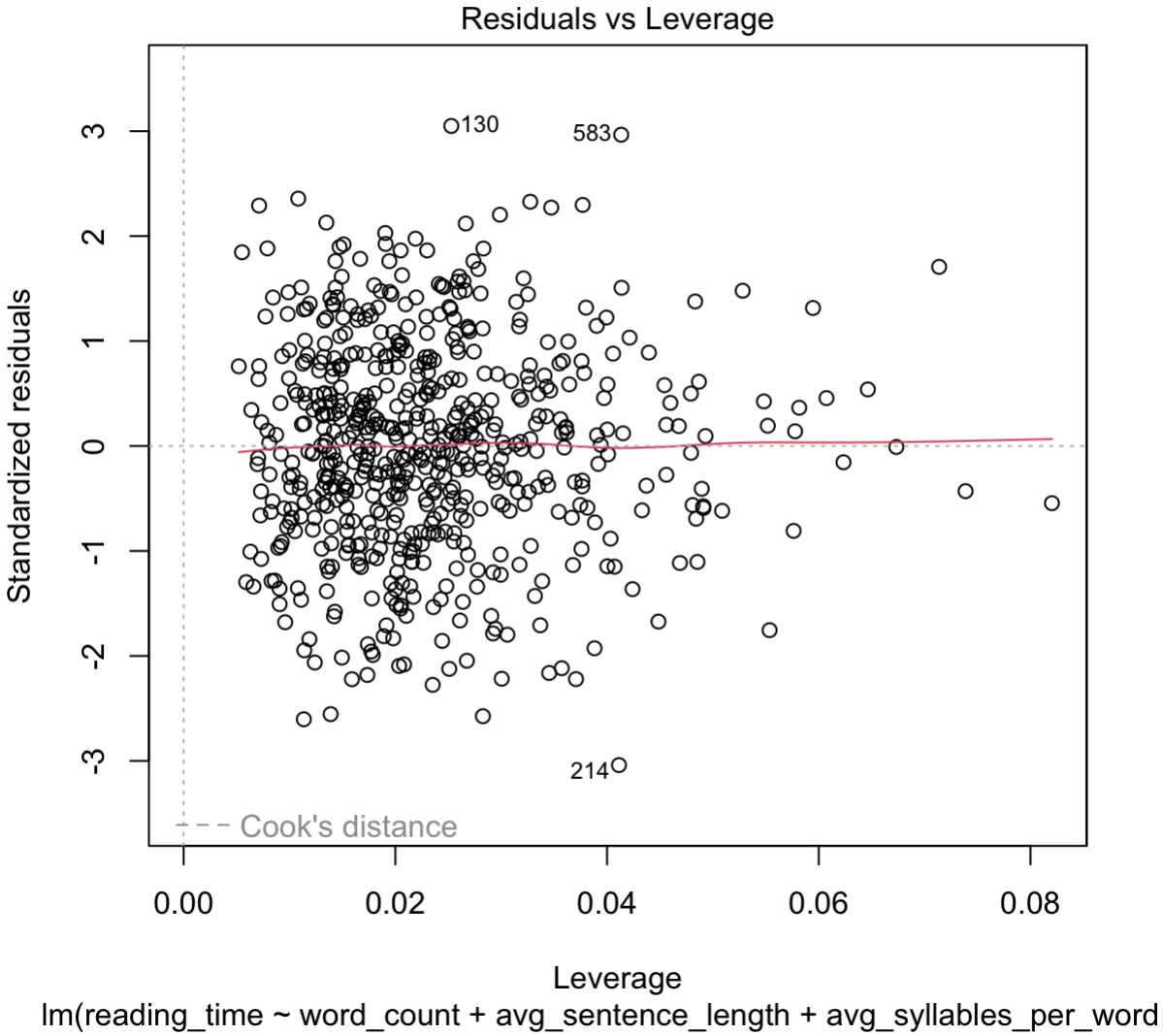
a) **Observations:** The points are distributed as a random "cloud" around the horizontal dashed line at zero. The red smoothing line is nearly flat and stays close to the zero axis across the entire range of fitted values (from approximately 5 to 22 minutes).
b) **Conclusion:** There is no evidence of non-linear patterns or heteroscedasticity (funnel shapes). This confirms that the model correctly captures the functional form of the data and that the standard errors of our parameters are reliable.



Normality of Residuals (Normal Q-Q Plot): The residuals align almost perfectly with the diagonal reference line. Only minor deviations are observed at the extreme ends (points 214, 583, 130), which do not significantly impact the model's reliability. This confirms that the errors follow a normal distribution.



Scale-Location: The red line in this plot is relatively flat, providing further evidence that the variance of the residuals is consistent across all levels of predicted reading time (homoscedasticity).



Residuals vs Leverage: This plot identifies influential data points. All observations fall within the Cook's distance boundaries, indicating that no single document disproportionately biases the model's coefficients.

Conclusion on Assumptions: All standard modeling assumptions are satisfied. The model is robust and suitable for high-confidence predictions.

5) Using the selected **minimal_model**, we calculated the predicted reading times and the intervals capturing the middle 90% of observations for a document with the specific structural and linguistic features provided by management.

Document Type	Predicted Mean (fit)	90% Lower Bound (lwr)	90% Upper Bound (upr)
(a) General	**11.27 min**	**9.55 min**	**13.00 min**
(b) Technical	**10.49 min**	**8.75 min**	**12.22 min**

The model provides a specific range within which we expect 90% of similar documents to fall:

Interestingly, for this specific set of characteristics (1300 words, readability 63.9, typo rate 0.05, etc.), the **General** document is predicted to take slightly longer to read (~11.27 min) than the **Technical** one (~10.49 min).

This outcome demonstrates the importance of **interaction terms** in our model. While technical documents are often harder, the specific combination of a high readability score (63.9 is relatively "easy") and a high typo rate (0.05) may penalize the general document more severely or benefit the technical document more significantly according to the fitted coefficients.

The manager can now state with 90% confidence that a general document with these features will be read in roughly 9.5 to 13 minutes, while a technical one will likely be faster, between 8.7 and 12.2 minutes.

Exercise 4)

Logistic Regression Analysis (HMDA)

To investigate the determinants of mortgage denial, a Logistic Regression (Logit) model was employed. The response variable deny is binary, making the logit link function appropriate for estimating probabilities within the [0, 1] interval.

Following a stepwise selection procedure based on the **Akaike Information Criterion (AIC)**, the final model retained 10 significant predictors: pirat, lvrat, chist, phist, unemp, selfemp, insurance, afam, single, and hschool.

The AIC for this model is **1292**, indicating an optimal balance between model complexity and goodness of fit.

While AIC included 10 variables, a BIC selection (not shown here but performed) typically results in a more parsimonious model by applying a stricter penalty for the large sample size (n=2380).

The coefficients provide insights into how financial and demographic factors affect the probability of mortgage denial :

Financial Ratios: Both the payment-to-income ratio (pirat, β~ 4.63) and loan-to-value ratio (lvrat, β~1.93) are highly significant and positive ($p < 0.001$). This confirms that higher debt-to-income and higher leverage significantly increase the risk of denial.

Credit History: All levels of consumer credit history (chist) and public bad credit records (phist, β~1.27) show significant positive coefficients, indicating that poor past credit performance is a primary driver for rejection.

Demographics: Being African-American (afamyes, β~ 0.69) is a significant predictor ($p < 0.001$), suggesting that race affects denial even when controlling for financial variables. Conversely, having a high school diploma (hschoolyes, β~ -1.04) significantly reduces the probability of denial.

The question specifically asks how being single affects the odds of denial compared to being married. According to the AIC-selected model:

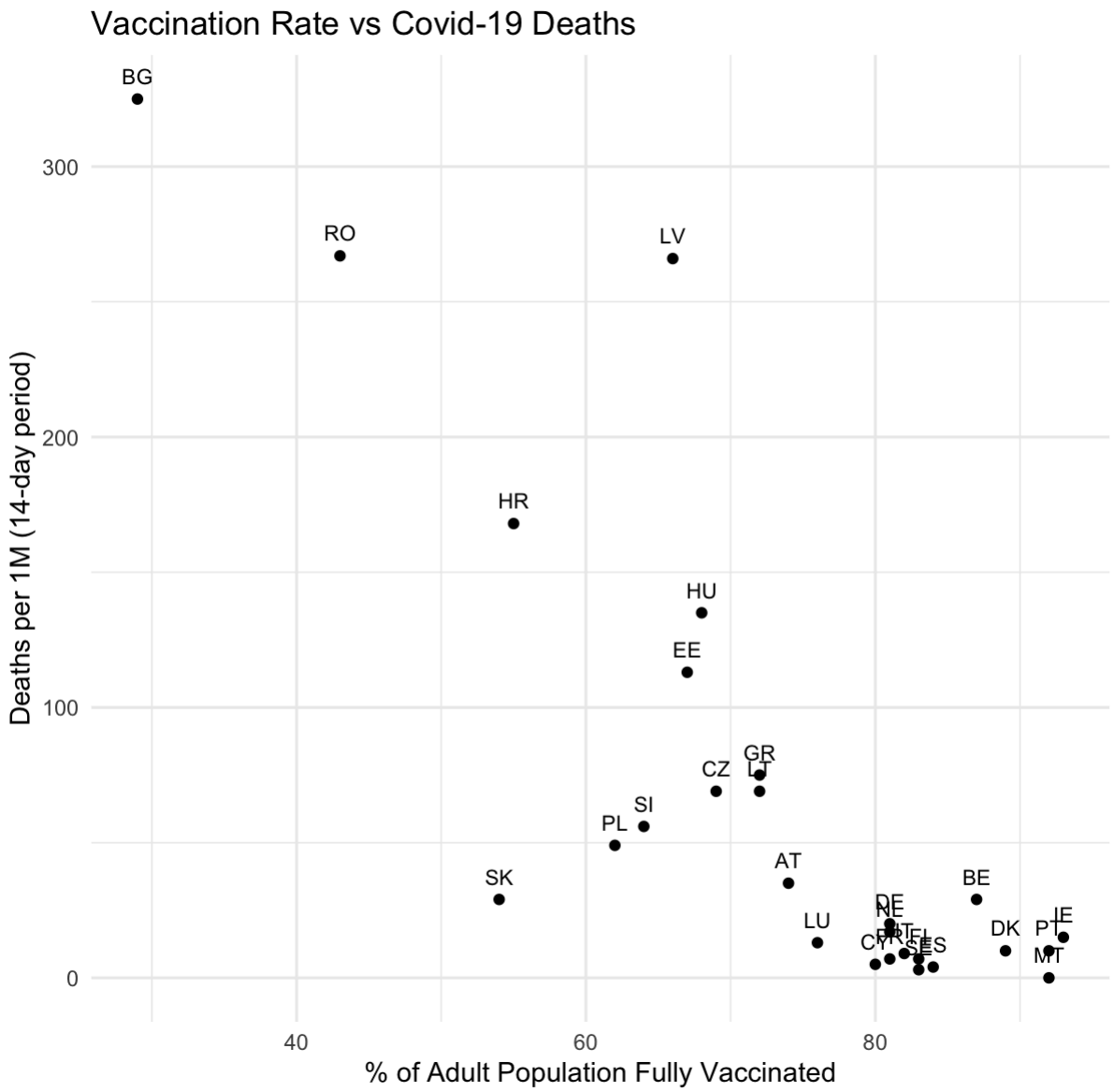
Coefficient (β): The coefficient for singleyes is **0.44167** and is statistically significant ($p = 0.00369$).

Odds Ratio (e^β): Calculating the exponentiated coefficient gives an Odds Ratio of **1.555**.

Holding all other financial and personal characteristics constant (such as income ratios and credit history), **singles have 1.555 times higher odds** of being denied a mortgage compared to married individuals. This represents a **55.5% increase** in the odds of denial for single applicants compared to their married counterparts with similar profiles.

Exercise 5)

To understand the relationship between the percentage of the adult population fully vaccinated and the number of deaths per 1 million inhabitants, we generated a scatter plot.



The plot reveals a **strong negative correlation** between vaccination rates and COVID-19 mortality. As the percentage of fully vaccinated adults increases, the number of deaths per million significantly decreases.

The relationship does not appear to be perfectly linear. Instead, it follows a **curved (exponential) decay** pattern. Countries with low vaccination rates (below 60%, such as Bulgaria - BG and Romania - RO) exhibit disproportionately higher death rates compared to countries that have surpassed the 80% vaccination threshold.

We can observe two distinct clusters. High-vaccination countries (IE, PT, MT, DK) are tightly grouped near the bottom-right of the plot with near-zero mortality, while low-vaccination countries are more widely scattered with much higher mortality rates.

Bulgaria (BG) stands out as an influential point with the lowest vaccination rate and the highest mortality rate in the dataset.

To analyze the effect of vaccination rates on deaths, we compared two distinct statistical approaches:

a) Simple Linear Regression (OLS): Used as a baseline model to estimate the constant average decrease in deaths for every unit increase in vaccination.

b) Poisson Regression (GLM): Specifically chosen because our response variable (number of deaths) represents count data. This model is more appropriate as it assumes a Poisson distribution, naturally handling non-negative integers and accounting for the non-linear relationship observed in the exploratory phase.

Below is the summary of the estimated parameters for both models:

Metric / Parameter	**Linear Model (OLS)**	**Poisson Model (Log-link)**
Intercept (β0)	421.00	7.528
Vax_rate Coeff (β1)	-4.8317	-0.0504
P-value (vax_rate)	1.74 x 10^{-7} (***)	< 2 x 10^{-16} (***)
Model Fit	Adjusted R^2: 0.658	AIC: 1047

Linear Perspective: According to the OLS model, every 1% increase in the adult vaccination rate is associated with an average decrease of **4.83 deaths** per million

Poisson Perspective: The Poisson model reveals a proportional effect. By calculating 1 - e^{-0.0504} ~ 0.049, we find that **every 1% increase in vaccination reduces the expected number of deaths by approximately 4.9%**.

In both models, the vaccination rate is a highly significant predictor ($p < 0.001$), confirming that higher vaccination coverage is robustly linked to lower mortality across the EU.

Explanation for Non-Experts :

We used two different mathematical methods to test how vaccines protect the population. Both methods tell the same story: vaccines are incredibly effective. Imagine two countries: if Country A increases its vaccination coverage by just 1% more than Country B, it can expect nearly a 5% reduction in its death rate. This isn't just a small improvement; it's a consistent, powerful trend across all 27 EU nations. Whether we look at it as a flat number of lives saved or a percentage drop, the conclusion is clear: higher vaccination rates are the primary reason some countries have almost zero deaths while others struggle with high mortality.

Limitations :

In the Poisson model, the Residual Deviance (906.3) is much larger than the degrees of freedom (25). This indicates overdispersion, suggesting that the actual variance in deaths is higher than the model assumes.

The analysis does not include variables like healthcare quality, age demographics, or the prevalence of specific virus variants, which also influence mortality rates.